AGRI BOX: An IoT-Based Offline Environmental Data Logger with Wireless Synchronization

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***Abstract*—This paper presents AGRI BOX, an IoT-based environmental monitoring system designed for deployment in remote areas lacking internet connectivity. The system uses a dual ESP32 architecture consisting of a Logger Station and a Base Station. The Logger collects environmental data such as temperature, humidity, pressure, soil moisture, and fall detection events, storing it locally using LittleFS. When a Base Station comes within range, data is transmitted wirelessly using WiFi and uploaded to a Firebase database for real-time visualization. The system operates in a store-and-forward model, ensuring reliability, low power consumption, and offline capability. A dynamic web dashboard provides real-time insights through graphical visualization.**

***Index Terms*—IoT, ESP32, Data Logger, Base, Wireless Synchronization, Firebase, Environmental Monitoring**

I. Introduction

Monitoring environmental conditions in remote agricultural or forest areas is a significant challenge due to lack of connectivity. Traditional IoT systems depend heavily on continuous internet access, making them unsuitable for such regions.

AGRI BOX addresses this limitation by introducing an offline-first architecture using a store-and-forward mechanism. The system ensures data reliability, energy efficiency, and ease of deployment without requiring constant connectivity.

II. Literature Overview

A. Overview of Existing IoT Applications Related to the Research Domain

Existing IoT applications in the agricultural and environmental monitoring domain mainly focus on continuous sensing, wireless data transfer, cloud integration, and dashboard-based visualization. In [1], a WiFi-based environmental data logger was developed for greenhouse monitoring. That system collected environmental parameters such as air temperature, air humidity, and water temperature using multiple sensors, and stored the collected measurements in a local database through a Raspberry Pi-based architecture. A web interface was also designed to visualize sensor outputs and support remote monitoring.

In [3], an ESP32-based real-time temperature and humidity monitoring system was proposed for weather observation. That work used a DHT11 sensor, RTC, and LCD interface, while data was uploaded to a PythonAnywhere cloud platform through HTTP. The server-side workflow used Flask and SQLite, allowing the sensed data to be stored, viewed remotely, and exported as CSV files for later analysis.

In [2], IoT was applied in smart agriculture through an ESP32-based wireless sensor network integrated with Blynk and MATLAB. The system targeted agricultural monitoring and automated irrigation, with emphasis on crop-related variables such as air temperature, air humidity, soil moisture, and pH level. More than 1,000 sensed data entries were analyzed to support precision agriculture and crop-condition assessment.

These prior works show that IoT has already been used effectively for greenhouse monitoring, real-time weather observation, and precision agriculture. However, most of them assume either fixed infrastructure or continuous data connectivity during operation.

B. Discussion of Previous Approaches, Their Limitations, and How the Proposed Work Differs/Improves Upon Them

The study in [1] demonstrated a multi-sensor environmental data logger with website-based monitoring. However, its architecture relied on a Raspberry Pi as the main coordinator and data storage unit. The paper explicitly noted the storage and memory limitations of ESP32 and therefore preferred Raspberry Pi with larger storage capacity for long-term logging. While suitable for greenhouse environments with available infrastructure, this approach is relatively heavier, more power-dependent, and better suited for fixed deployments than for highly mobile or disconnected field scenarios.

The work in [3] used a simpler ESP32-centered architecture and successfully implemented cloud-based weather logging. Nevertheless, its design still depended on continuous HTTP communication with the PythonAnywhere platform, together with Flask and SQLite for server-side storage. Since the system was designed around direct online uploading, it is less suitable for remote farms or forest areas where internet access is poor or unavailable. In addition, its sensing scope was limited mainly to temperature and humidity.

The work in [2] advanced agricultural IoT by combining wireless sensing with data analytics through Blynk and MATLAB. It demonstrated the value of large-scale data collection and analysis for crop monitoring and irrigation-related decisions. However, this system also depended on a centralized online platform and real-time accessibility of sensed data. Its main focus was on precision agriculture analytics rather than on resilient operation in disconnected environments.

Compared with these approaches, the proposed AGRI BOX system is designed specifically for connectivity-poor "dead zones." Instead of requiring always-on internet or a continuously connected gateway, AGRI BOX adopts an offline-first store-and-forward model. The Logger Station collects environmental data and stores it locally in LittleFS, while a mobile Base Station later retrieves the stored records wirelessly and uploads them to Firebase. This makes the proposed system better suited for remote agricultural or forest deployments where continuous connectivity cannot be guaranteed. In addition, the use of a dual-ESP32 architecture reduces system weight and complexity compared to Raspberry Pi-based approaches, while still enabling later cloud synchronization and dashboard visualization.

C. Summary of Key Research Gaps

From the reviewed literature, three major research gaps can be identified.

First, most existing systems are designed around continuous or near-real-time connectivity. Whether the architecture uses router-based WiFi, cloud-hosted platforms, or centralized online dashboards, the basic assumption is that network access is available during sensing and logging. This creates a gap for applications in remote agricultural areas where such connectivity does not exist.

Second, previous studies place greater emphasis on real-time viewing, cloud upload, and analytical processing than on robust offline data preservation. Although local or server-side databases are used, the reviewed works do not center their architecture on delayed synchronization between an autonomous field node and a mobile retrieval node. As a result, they do not directly address the problem of reliable environmental monitoring in disconnected zones.

Third, the sensing and operational objectives of prior systems are comparatively narrow. Some focus mainly on greenhouse parameters, others on weather monitoring, and others on agricultural analytics. There remains a gap in lightweight field-deployable systems that combine environmental sensing, low-power operation, offline logging, wireless synchronization, and practical deployment feedback features in one integrated design.

Therefore, the main gap addressed by the proposed work is the lack of a lightweight, low-power, offline-capable IoT monitoring system that can autonomously log environmental data in remote areas and later synchronize that data wirelessly to a mobile gateway for database upload and dashboard visualization.

III. System Overview

The AGRI BOX system consists of two main components:

- **Logger Station**: Collects and stores environmental data.
- **Base Station**: Retrieves stored data and uploads it to a cloud database.

The communication follows a master-slave topology where the Base Station initiates synchronization.

IV. Hardware Architecture

A. Logger Station

The Logger Station is built around an ESP32 microcontroller and is deployed in the field.

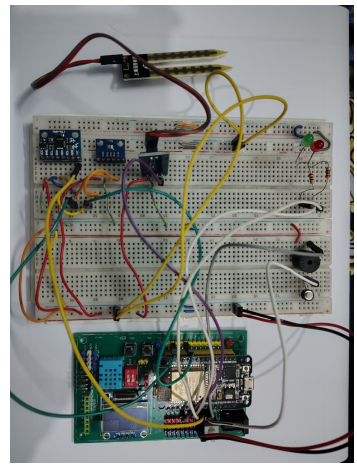

Fig. 1. Logger Station (Top View)

It includes the following sensors:

- BMP280 (Temperature and Pressure)
- DHT11 (Humidity)
- Soil Moisture Sensor
- GY-521 (Fall Detection)

Additional components:

- OLED Display
- LEDs (Status Indicators)
- Buzzer

This is the Logger Station wiring plan used in this project,

TABLE I
Logger Station Wiring

Component Signal	ESP32 GPIO	Notes
BMP280 SDA	21	I2C shared
BMP280 SCL	22	I2C shared
GY-521 SDA	21	I2C shared
GY-521 SCL	22	I2C shared
DHT11 Digital	04	Digital input
OLED SDA	21	I2C shared
OLED SCL	22	I2C shared
Soil Sensor AO	34	Analog input
Green LED (+)	32	220Ω resistor
Red LED (+)	33	220Ω resistor
Buzzer (+)	27	Active buzzer
Fall Reset Button	25	Active Low

B. Base Station

The Base Station is also powered by an ESP32 and acts as a mobile gateway.

It is responsible for:

- Broadcasting connection signals
- Receiving data packets
- Uploading data to Firebase

This is the Logger Station wiring plan used in this project,

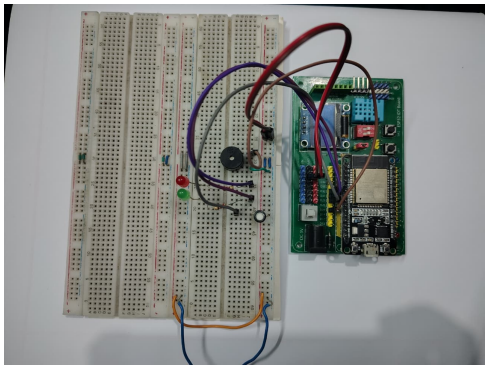

Fig. 2. Base Station (Top View)

TABLE II
BASE STATION WIRING

Component	ESP32 GPIO	Notes
OLED SDA	21	I2C shared
OLED SCL	22	I2C shared
Green LED	32	220Ω resistor
Red LED	33	220Ω resistor
Buzzer	27	Active buzzer
Fall Reset Button	25	Active Low

V. SYSTEM OPERATION

A. Offline Logging Phase

The Logger wakes from deep sleep, reads sensor values, formats them into CSV, and stores them in internal flash memory (LittleFS). This allows continuous operation without internet.

B. Synchronization Phase

When the Base Station is nearby:

- Base sends a "PING"
- Logger responds with "READY"
- Data transmission begins

C. Data Transfer

Data is transmitted in packets via WiFi. After successful transfer:

- Base confirms receipt
- Logger deletes stored data

VI. SOFTWARE ARCHITECTURE

A. Embedded System

The system is programmed using Arduino IDE with ESP32 libraries. Key functionalities include:

- Sensor data acquisition
- File handling using LittleFS
- WiFi communication protocol
- Power management via deep sleep

Fig. 3. Web Dashboard Visualization

B. Web Dashboard

A dynamic web interface displays real-time data from Firebase.

The dashboard:

- Displays graphs and timestamps
- Updates automatically with new data

VII. USER INTERFACE AND FEEDBACK

A. LED Indicators

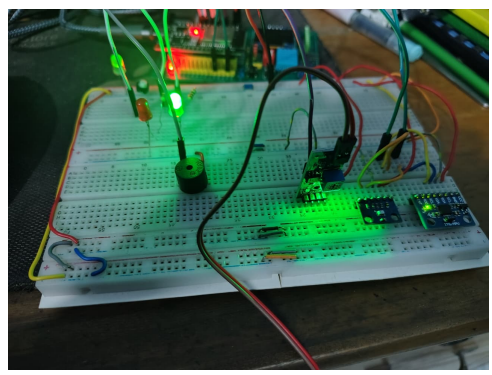

Fig. 4. Green LED (Normal Operation)

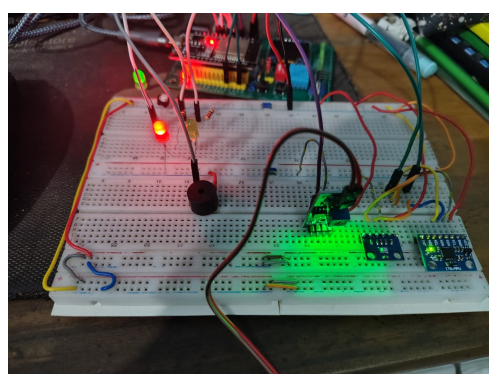

Fig. 5. Red LED (Error/Alert State)

- Green LED: Normal operation and successful sync
- Red LED: Errors such as WiFi failure or storage issues

B. Buzzer and OLED

- The buzzer provides auditory feedback for events
- The OLED displays the system status (OK, FALL DETECTED, STORAGE FULL)

VIII. Fall Detection Mechanism

The GY-521 sensor detects sudden movement or falls. Upon detection:

- Logging is paused
- Alert is sent to Base Station
- LED and buzzer indicate emergency

A reset button restores normal operation.

IX. Results and Performance

The system successfully demonstrates:

- Reliable offline data logging
- Efficient wireless synchronization
- Real-time cloud integration
- Low power consumption

X. Conclusion

AGRI BOX provides a robust solution for environmental monitoring in remote areas. Its store-and-forward architecture ensures reliability without continuous connectivity. The integration of IoT hardware, wireless communication, and cloud visualization makes it highly practical for agricultural and environmental applications.

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